

Equivalence of formulas: Let A and B be two statement formulas and let $P_1, P_2, P_3, \dots, P_n$ denote all the variables occurring in both A and B. If the truth value of A is equal to the truth value of B for every 2^n possible combination of truth values, then A and B are said to be equivalent and it is denoted by $A \Leftrightarrow B$

Example: $\sim\sim P \Leftrightarrow P$, $P \vee P$, etc.

Remarks:

1. $\Leftrightarrow$ is not a connective.
2. Two statement formulas A and B are equivalent iff $A \leftrightarrow B$ is a tautology.

Example : Prove that

(a) $P \rightarrow Q \Leftrightarrow \sim P \vee Q,$

(b) $P \leftrightarrow Q \Leftrightarrow (P \rightarrow Q) \wedge (Q \rightarrow P)$

(c) $\sim (P \wedge Q) \Leftrightarrow \sim P \vee \sim Q$

(d) $\sim (P \vee Q) \Leftrightarrow \sim P \wedge \sim Q$

Equivalent formulae:

1. $P \vee P \Leftrightarrow P$ and $P \wedge P \Leftrightarrow P$ (Idempotent law)
2. $(P \vee Q) \vee R \Leftrightarrow P \vee (Q \vee R)$ and $(P \wedge Q) \wedge R \Leftrightarrow P \wedge (Q \wedge R)$ (Associative law)
3. $P \vee Q \Leftrightarrow Q \vee P$ and $P \wedge Q \Leftrightarrow Q \wedge P$ (Commutative law)
4. $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$ and $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$ (Distributive law)
5. $P \vee F \Leftrightarrow P$ and $P \wedge T \Leftrightarrow P$ (Identity law)
6. $P \vee T \Leftrightarrow T$ and $P \wedge F \Leftrightarrow F$

7. $P \vee \sim P \Leftrightarrow T$ and $P \wedge \sim P \Leftrightarrow F$

(Negation law)

8. $P \vee (P \wedge Q) \Leftrightarrow P$ and $P \wedge (P \vee Q) \Leftrightarrow P$

(Absorption law)

9. $\sim(P \vee Q) \Leftrightarrow \sim P \wedge \sim Q$ and $\sim(P \wedge Q) \Leftrightarrow \sim P \vee \sim Q$

(Demorgan's law)

Logic equivalence involving implications

Implications
$p \rightarrow q \Leftrightarrow \neg p \vee q$
$p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$
$\neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q$
$p \vee q \Leftrightarrow \neg p \rightarrow q$
$p \wedge q \Leftrightarrow \neg(p \rightarrow \neg q)$
$(p \rightarrow q) \wedge (p \rightarrow r) \Leftrightarrow p \rightarrow (q \wedge r)$
$(p \rightarrow q) \vee (p \rightarrow r) \Leftrightarrow p \rightarrow (q \vee r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \Leftrightarrow (p \vee q) \rightarrow r$
$(p \rightarrow r) \vee (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$

Logic equivalence involving Bi conditional

Biconditions
$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \Leftrightarrow \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \Leftrightarrow p \leftrightarrow \neg q$

Without using Truth tables, prove the following

(i) Show that $(p \vee q) \wedge \neg p \Leftrightarrow (\neg p \wedge q)$

(ii) Show that $(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$

(iii) Show that $\neg(p \vee (\neg p \wedge q)) \Leftrightarrow \neg p \wedge \neg q$

(iv) Show that $\neg(\neg((p \vee q) \wedge r) \vee \neg q) \Leftrightarrow q \wedge r$

(v) Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

(vi) Show that $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$.

(i)

$S \Leftrightarrow (p \vee q) \wedge \neg p$	Reasons
$\Leftrightarrow (p \vee q) \wedge \neg p$	Given
$\Leftrightarrow (p \wedge \neg p) \vee (q \wedge \neg p)$	Distributive law
$\Leftrightarrow F \vee (\neg p \wedge q)$	Negation law, Commutative law
$\Leftrightarrow \neg p \wedge q$	Identity law

(ii)

$S \Leftrightarrow (p \vee q) \wedge \neg(\neg p \wedge q)$	Reasons
$\Leftrightarrow (p \vee q) \wedge \neg(\neg p \wedge q)$	Given
$\Leftrightarrow (p \vee q) \wedge (\neg \neg p \vee \neg q)$	De Morgan's law
$\Leftrightarrow (p \vee q) \wedge (p \vee \neg q)$	Negation law
$\Leftrightarrow p \vee (q \wedge \neg q)$	Distributive law
$\Leftrightarrow p \vee F$	Negation law
$\Leftrightarrow p$	Identity law

(iii)

$S \Leftrightarrow \neg(p \vee (\neg p \wedge q))$	Reasons
$\Leftrightarrow \neg(p \vee (\neg p \wedge q))$	Given
$\Leftrightarrow \neg p \wedge \neg(\neg p \wedge q)$	De Morgan's law
$\Leftrightarrow \neg p \wedge (p \vee \neg q)$	De Morgan's law
$\Leftrightarrow (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	Distributive law
$\Leftrightarrow F \vee (\neg p \wedge \neg q)$	Negation law
$\Leftrightarrow \neg p \wedge \neg q$	Identity law

(iv)

$S \Leftrightarrow \neg(\neg((p \vee q) \wedge r) \vee \neg q) \Leftrightarrow q \wedge r$	Reasons
$\Leftrightarrow \neg(\neg((p \vee q) \wedge r) \vee \neg q)$	Given
$\Leftrightarrow ((p \vee q) \wedge r) \wedge q$	De Morgan's law
$\Leftrightarrow (p \vee q) \wedge (r \wedge q)$	Associative law
$\Leftrightarrow (p \wedge (r \wedge q)) \vee (q \wedge (r \wedge q))$	Distributive law
$\Leftrightarrow (p \wedge (r \wedge q)) \vee (r \wedge q)$	Idempotent law
$\Leftrightarrow r \wedge q$	Absorption law

Tautological Implications:

A statement A is said to be tautologically imply a statement B iff $A \rightarrow B$ is a tautology and it is denoted by $A \Rightarrow B$, which is read as A implies B.

Remark:

1. ' $\Rightarrow$ ' is not a connective.
2. $A \Rightarrow B$ is not a statement formula.
3. ' $\Rightarrow$ ' is transitive.
4. If $H_1, H_2, \dots, H_n \Rightarrow Q$, then $(H_1 \wedge H_2 \wedge \dots \wedge H_n \Rightarrow Q)$

Example:

1. Prove that $P \wedge (P \rightarrow Q) \Rightarrow Q$ using truth table and without using truth table.
2. $(P \wedge Q) \Rightarrow (P \rightarrow Q)$
3. $\sim(P \leftrightarrow Q) \Leftrightarrow (P \vee Q) \wedge \sim(P \wedge Q)$